# Old Exam Questions
# Unsupervised Machine Learning

## Worked Solutions

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**How this file is organized**

This is the **solutions companion** to `exams.tex`. Every question is followed *immediately* by a green **answer box**. For **multiple-choice** questions each option is marked ✓ (correct) or × (incorrect), with a short justification — *why* the correct options are correct and, for the tricky / near-miss distractors, *why* they are wrong. For **calculations** the working is shown. Options are referred to by position (A = first option, B = second, . . . ). If you want a clean practice set with blank answers, use `exams.tex`.

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# Probability, Information Theory, and Estimation

**Question 1: Cumulants for independent uniform variables**

Assume you have two independent random variables $X$ and $Y$ which are uniformly distributed on $[-1, 1]$. As in the lecture, let $\kappa_n(X)$ denote the $n$-th order cumulant of a random variable $X$. Compute the following quantities and answer the last classification question. Use the hint

$$\kappa_4(X) = \mathbb{E}(X^4) - 3\big(\mathbb{E}(X^2)\big)^2$$

for centered $X$.

- $\kappa_4(X)$
- $\kappa_4(2X)$
- $\kappa_4(X + Y)$
- $X$ is sub-Gaussian / super-Gaussian

**Answer**

For a discrete random variable, an expectation is a probability-weighted sum; for a continuous random variable with density $f_X$, it is an integral:

$$\mathbb{E}[g(X)] = \sum_x g(x)P(X = x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)\,\mathrm{d}x.$$

To compute a moment, choose $g(x) = x^r$. Since $X \sim U[-1, 1]$, its density is

$$f_X(x) = \frac{1}{2} \quad \text{for} \quad -1 \le x \le 1 \qquad \text{and} \qquad f_X(x) = 0 \quad \text{otherwise.}$$

Consequently,

$$\mathbb{E}(X^r) = \int_{-1}^{1} x^r \frac{1}{2}\,\mathrm{d}x.$$

Odd moments are zero because an odd integrand cancels on the symmetric interval. For even moments $r = 2k$,

$$\mathbb{E}(X^{2k}) = \frac{1}{2}\left[\frac{x^{2k+1}}{2k+1}\right]_{-1}^{1} = \frac{1}{2k+1}.$$

In particular,

$$\mathbb{E}(X^2) = \frac{1}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{1}{3}, \qquad \mathbb{E}(X^4) = \frac{1}{2}\left[\frac{x^5}{5}\right]_{-1}^{1} = \frac{1}{5}.$$

Also $\mathbb{E}(X) = 0$ by symmetry, so $X$ is centered and the given cumulant formula applies:

$$\kappa_4(X) = \tfrac{1}{5} - 3\big(\tfrac{1}{3}\big)^2 = \tfrac{1}{5} - \tfrac{1}{3} = -\tfrac{2}{15} \approx \mathbf{-0.13}.$$

Cumulants of order $n$ scale as $\kappa_n(cX) = c^n\kappa_n(X)$. Therefore,

$$\kappa_4(2X) = 2^4\kappa_4(X) = 16\left(-\frac{2}{15}\right) = -\frac{32}{15} \approx \mathbf{-2.13}.$$

For independent variables, cumulants of the same order add. Since $X$ and $Y$ have the same distribution,

$$\kappa_4(X + Y) = \kappa_4(X) + \kappa_4(Y) = 2\left(-\frac{2}{15}\right) = -\frac{4}{15} \approx \mathbf{-0.27}.$$

Since $\kappa_4(X) < 0$ (negative excess kurtosis), $X$ is **sub-Gaussian** — the uniform distribution has lighter tails than a Gaussian.

**Question 2: Cumulants for symmetric Bernoulli variables**

Assume you have two independent, discrete random variables $X$ and $Y$, both following independent symmetric Bernoulli distributions taking only the values $+1$ and $-1$, each with probability $1/2$. As in the lecture, let $\kappa_n(X)$ denote the $n$-th order cumulant of a random variable $X$. Compute the quantities below. Use the hint

$$\kappa_3(X) = \mathbb{E}(X^3), \qquad \kappa_4(X) = \mathbb{E}(X^4) - 3\big(\mathbb{E}(X^2)\big)^2$$

for centered $X$.

- $\kappa_3(X)$
- $\kappa_4(X)$
- $\kappa_4(3X)/\kappa_4(X)$
- $\kappa_4(X+Y)/\kappa_4(X)$

**Answer**

For a discrete random variable,

$$\mathbb{E}[g(X)] = \sum_x g(x)P(X = x).$$

Here the only possible values are $-1$ and $1$, each with probability $\frac{1}{2}$. Thus,

$$\mathbb{E}(X) = (-1)\frac{1}{2} + (1)\frac{1}{2} = 0,$$

$$\mathbb{E}(X^2) = (-1)^2\frac{1}{2} + 1^2\frac{1}{2} = 1, \qquad \mathbb{E}(X^3) = (-1)^3\frac{1}{2} + 1^3\frac{1}{2} = 0,$$

$$\mathbb{E}(X^4) = (-1)^4\frac{1}{2} + 1^4\frac{1}{2} = 1.$$

Equivalently, $X^2 = X^4 = 1$ always and $X^3 = X$. Because $\mathbb{E}(X) = 0$, $X$ is centered, so

$$\kappa_3(X) = \mathbb{E}(X^3) = \mathbb{E}(X) = \mathbf{0} \qquad \text{(the distribution is symmetric)},$$

$$\kappa_4(X) = \mathbb{E}(X^4) - 3\big(\mathbb{E}(X^2)\big)^2 = 1 - 3 = \mathbf{-2}.$$

Using homogeneity of fourth-order cumulants,

$$\frac{\kappa_4(3X)}{\kappa_4(X)} = \frac{3^4\kappa_4(X)}{\kappa_4(X)} = 3^4 = \mathbf{81}.$$

Using additivity for the independent variables $X$ and $Y$ and the fact that they are identically distributed,

$$\frac{\kappa_4(X+Y)}{\kappa_4(X)} = \frac{\kappa_4(X) + \kappa_4(Y)}{\kappa_4(X)} = \frac{2\kappa_4(X)}{\kappa_4(X)} = \mathbf{2}.$$

**Question 3: Eigenvectors, covariance matrix, and a scalar expression**

You are given a dataset with $n$ samples and three features. The covariance matrix $C$ has eigenvalues $(\lambda_1, \lambda_2, \lambda_3) = (5, 3, 1)$ and associated normalized eigenvectors $(v_1, v_2, v_3)$ with $\|v_i\| = 1$. Compute

$$A = (2v_1^\top + v_2^\top + 3v_3^\top)\, C\, (3v_2 + v_3).$$

**Answer**

For a real symmetric covariance matrix, eigenvectors belonging to different eigenvalues are orthogonal. Together with the given normalization this means $v_i^\top v_j = \delta_{ij}$: the inner product is 1 when $i = j$ and 0 otherwise. Also, by the definition of an eigenvector, $Cv_i = \lambda_i v_i$. First multiply from the right:

$$C(3v_2 + v_3) = 3Cv_2 + Cv_3 = 3\lambda_2 v_2 + \lambda_3 v_3 = 9v_2 + v_3.$$

Now expand all products:

$$\begin{aligned} A &= (2v_1^\top + v_2^\top + 3v_3^\top)(9v_2 + v_3) \\ &= 18\, v_1^\top v_2 + 2\, v_1^\top v_3 + 9\, v_2^\top v_2 + v_2^\top v_3 + 27\, v_3^\top v_2 + 3\, v_3^\top v_3 \\ &= 0 + 0 + 9 + 0 + 0 + 3 = \mathbf{12}. \end{aligned}$$

## Question 4: KL divergence for continuous uniform distributions

*Asked multiple times.*

Assume $Y$ and $Z$ are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively:

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y}, & x \in [0, \theta_Y], \\ 0, & \text{else}, \end{cases} \qquad p_Z(x) = \begin{cases} \frac{1}{\theta_Z}, & x \in [0, \theta_Z], \\ 0, & \text{else}. \end{cases}$$

For $\theta_Y = e^5$ and $\theta_Z = e^{10}$, compute

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log\left(\frac{p_Y(x)}{p_Z(x)}\right) \mathrm{d}x.$$

Use the natural logarithm and round to two decimal places.

**Answer**

In general, the integral only has to cover the support of the first distribution $p_Y$, because the factor $p_Y(x)$ is zero elsewhere. Here this support is $[0, \theta_Y]$, and it lies inside $[0, \theta_Z]$ because $e^5 < e^{10}$. On this interval both densities are nonzero and their ratio is constant:

$$\begin{aligned} D_{\mathrm{KL}}(p_Y \| p_Z) &= \int_0^{\theta_Y} \frac{1}{\theta_Y} \ln\left(\frac{1/\theta_Y}{1/\theta_Z}\right) \mathrm{d}x \\ &= \ln\left(\frac{\theta_Z}{\theta_Y}\right) \underbrace{\int_0^{\theta_Y} \frac{1}{\theta_Y} \mathrm{d}x}_{=1} \\ &= \ln\left(\frac{e^{10}}{e^5}\right) = \ln(e^5) = \mathbf{5.00}. \end{aligned}$$

(This works because the support of $p_Y$ is contained in the support of $p_Z$; the reverse divergence would be infinite.)

### Question 5: Outer product matrix

Given two vectors

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad y = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix},$$

compute the outer product matrix $A = xy^\top$ and give the third row of $A$.

### Answer

An outer product multiplies every component of the column vector $x$ with every component of the row vector $y^\top$. In symbols, $(xy^\top)_{ij} = x_i y_j$. Therefore,

$$A = xy^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1\cdot 4 & 1\cdot 5 & 1\cdot 6 \\ 2\cdot 4 & 2\cdot 5 & 2\cdot 6 \\ 3\cdot 4 & 3\cdot 5 & 3\cdot 6 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 8 & 10 & 12 \\ 12 & 15 & 18 \end{pmatrix}.$$

Hence the requested third row is

$$(\mathbf{12}, \mathbf{15}, \mathbf{18}).$$

### Question 6: Marginal probabilities, entropies, and mutual information — variant 1

*Asked multiple times.*

Assume you are given two binary random variables $X$ and $Y$ taking only the values 0 and 1 where their joint probability distribution is specified by

| | $Y=0$ | $Y=1$ |
|---|---|---|
| $X=0$ | 0.25 | 0.125 |
| $X=1$ | 0.25 | 0.375 |

Compute $P(X=1)$, $P(Y=1)$, $H(X)$, $H(Y)$, $I(X;Y)$ using the natural logarithm. Round the final values to two decimal places.

### Answer

Marginal probabilities are obtained by summing the joint distribution over the other variable:

$$P(X=x) = \sum_y P(X=x, Y=y), \qquad P(Y=y) = \sum_x P(X=x, Y=y).$$

Thus the complete marginals are

$$P(X=0) = 0.25 + 0.125 = 0.375, \qquad P(X=1) = 0.25 + 0.375 = 0.625 \approx \mathbf{0.63},$$

$$P(Y=0) = 0.25 + 0.25 = 0.50, \qquad P(Y=1) = 0.125 + 0.375 = \mathbf{0.50}.$$

For a discrete random variable, $H(X) = -\sum_x p_X(x) \ln p_X(x)$. Therefore,

$$\begin{aligned} H(X) &= -\big(0.375 \ln 0.375 + 0.625 \ln 0.625\big) \\ &\approx -\big(0.375(-0.981) + 0.625(-0.470)\big) \approx \mathbf{0.66}, \end{aligned}$$

$$H(Y) = -\big(0.5 \ln 0.5 + 0.5 \ln 0.5\big) = \ln 2 \approx \mathbf{0.69}.$$

The joint entropy uses all four cells:

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y)\ln p(x,y) \\ &= -\big(0.25\ln 0.25 + 0.125\ln 0.125 + 0.25\ln 0.25 + 0.375\ln 0.375\big) \\ &\approx 1.321. \end{aligned}$$

Finally, mutual information is the amount by which knowing one variable reduces the uncertainty about the other:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) \approx 0.662 + 0.693 - 1.321 \approx \mathbf{0.03}.$$

## Question 7: Marginal probabilities, entropies, and mutual information — variant 2

*Asked multiple times.*

Assume you are given two binary random variables $X$ and $Y$ taking only the values 0 and 1 where their joint probability distribution is specified by

| | $Y=0$ | $Y=1$ |
|---|---|---|
| $X=0$ | $\frac{1}{3}$ | $\frac{1}{3}$ |
| $X=1$ | 0 | $\frac{1}{3}$ |

Compute $P(Y=0)$, $H(X)$, $H(Y)$, $I(X;Y)$ using the natural logarithm. Round the final values to two decimal places.

### Answer

Add rows or columns of the joint table to obtain the marginals:

$$P(X=0) = \tfrac{1}{3}+\tfrac{1}{3} = \tfrac{2}{3}, \qquad P(X=1) = 0+\tfrac{1}{3} = \tfrac{1}{3},$$

$$P(Y=0) = \tfrac{1}{3}+0 = \tfrac{1}{3} \approx \mathbf{0.33}, \qquad P(Y=1) = \tfrac{1}{3}+\tfrac{1}{3} = \tfrac{2}{3}.$$

Using $H(X) = -\sum_x p_X(x)\ln p_X(x)$, both variables have the same entropy because both have marginal probabilities $\{\frac{1}{3},\frac{2}{3}\}$:

$$H(X) = H(Y) = -\left(\tfrac{2}{3}\ln\tfrac{2}{3} + \tfrac{1}{3}\ln\tfrac{1}{3}\right) \approx \mathbf{0.64}.$$

The joint distribution has three nonzero cells, each with probability $\frac{1}{3}$. The zero cell contributes $0\ln 0 := 0$, so

$$H(X,Y) = -3\left(\tfrac{1}{3}\ln\tfrac{1}{3}\right) = \ln 3 \approx 1.099.$$

Therefore,

$$I(X;Y) = H(X) + H(Y) - H(X,Y) \approx 0.637 + 0.637 - 1.099 \approx \mathbf{0.17}.$$

## Question 8: Marginal probabilities, entropies, and mutual information — variant 3

*Asked multiple times.*

Assume you are given two binary random variables $X$ and $Y$ taking only the values 0 and 1

where their joint probability distribution is specified by

| | $Y=0$ | $Y=1$ |
|---|---|---|
| $X=0$ | $\frac{3}{10}$ | $\frac{4}{10}$ |
| $X=1$ | $\frac{2}{10}$ | $\frac{1}{10}$ |

Compute $P(X=0)$, $P(Y=0)$, $H(X)$, $H(Y)$, $I(X;Y)$ using the natural logarithm. Round the final values to two decimal places.

**Answer**

Summing the rows and columns gives all marginal probabilities:

$$P(X=0) = 0.3 + 0.4 = \mathbf{0.70}, \qquad P(X=1) = 0.2 + 0.1 = 0.30,$$

$$P(Y=0) = 0.3 + 0.2 = \mathbf{0.50}, \qquad P(Y=1) = 0.4 + 0.1 = 0.50.$$

Insert these marginals into $H(X) = -\sum_x p_X(x) \ln p_X(x)$:

$$H(X) = -(0.7 \ln 0.7 + 0.3 \ln 0.3) \approx \mathbf{0.61}, \qquad H(Y) = -2(0.5 \ln 0.5) = \ln 2 \approx \mathbf{0.69}.$$

For the joint entropy, use the four entries of the original table:

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y) \ln p(x,y) \\ &= -\big(0.3 \ln 0.3 + 0.4 \ln 0.4 + 0.2 \ln 0.2 + 0.1 \ln 0.1\big) \\ &\approx 1.280. \end{aligned}$$

Hence,

$$I(X;Y) = H(X) + H(Y) - H(X,Y) \approx 0.611 + 0.693 - 1.280 \approx \mathbf{0.02}.$$

**Question 9: Parameter Estimation and Maximum Likelihood**

*Asked multiple times.*

Which statements about Parameter Estimation and Maximum Likelihood are true? Multiple answers are possible.

- ☐ If the data are distributed i.i.d., the likelihood can be written as a product of individual probabilities over the data samples.
- ☐ The likelihood and the log-likelihood are maximized for the same parameter value, and at that parameter value $\hat{w}$, their values coincide: $\mathcal{L}(\hat{w}) = \ln \mathcal{L}(\hat{w})$.
- ☐ Parameter estimation and the maximum likelihood approach rest on the assumption that the model class is not known and estimated from the data.
- ☐ The maximum likelihood estimator is asymptotically unbiased and efficient, but not asymptotically consistent.
- ☐ The mean squared error can be written as the sum of the variance of the estimator and the squared bias of the estimator.
- ☐ An estimator $\hat{w}$ is unbiased if its expected value equals the true parameter: $\mathbb{E}(\hat{w}) = w$.
- ☐ The likelihood $\mathcal{L}(\{x\}; w)$ is the probability that the model with parameter $w$ produces the data $\{x\}$.

☐ The mean squared error is defined as $\mathrm{mse}(\hat{w}) = (\hat{w} - w)^\top (\hat{w} - w)$.

**Answer**

✓ **A.** Independence lets the joint density factorize: $\mathcal{L} = \prod_i p(x_i; w)$.

× **B.** Same *argmax* (the logarithm is strictly monotone), but the *values* differ: $\ln \mathcal{L}(\hat{w}) \neq \mathcal{L}(\hat{w})$ in general.

× **C.** The model class is assumed to be *known and fixed*; only its parameters are estimated from the data.

× **D.** ML *is* asymptotically consistent (and unbiased and efficient) — the "not consistent" spoils the statement.

✓ **E.** Bias–variance decomposition: $\mathrm{mse}(\hat{w}) = \mathrm{Var}(\hat{w}) + \mathrm{bias}^2(\hat{w})$.

✓ **F.** That is the definition of unbiasedness.

✓ **G.** This matches the lecture's definition verbatim: the likelihood $\mathcal{L}(\{x\}; w) = p(\{x\}; w)$ is "the probability that the model produces the observed data." (Measure-theoretic caveat: for continuous data this is a *density*, not a probability — $P(X = x) = 0$ pointwise. The exam grades by the lecture's wording, so tick it.)

× **H.** The expectation is missing: $\mathrm{mse}(\hat{w}) = \mathbb{E}\big[(\hat{w} - w)^\top (\hat{w} - w)\big]$; without it, the expression is a random variable, not an error measure.

## Question 10: Dropdown selection task on estimation theory

*Asked multiple times.*

Fill in the correct choices.

For any parameter $w$, an estimator $\hat{w}$ is *consistent / efficient / unbiased* if $\mathbb{E}(\hat{w}) = w$. The *expected value / bias / mean squared error / variance* of $\hat{w}$ can be decomposed as a sum of its *expected value / bias / mean squared error / variance* and its squared *expected value / bias / mean squared error / variance*.

Let $X_1, \ldots, X_n$ be random variables drawn independently from a distribution $p(\cdot; w)$. Consider

$$\frac{1}{n+1} \sum_{i=1}^{n} X_i$$

as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is *neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent*.

Now consider $X_1$ as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is *neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent*.

Let $\{x_1, \ldots, x_n\}$ be a sample drawn independently from the distribution $p(\cdot; w)$. The function

$$\prod_{i=1}^{n} p(x_i; w)$$

is called the *probability / loss / density / likelihood* function. Since $p(\cdot; w)$ is usually a *probability / loss / density / likelihood* function, the probability measure of the set $\{x_1, \ldots, x_n\}$ is in fact zero.

The Cramér–Rao lower bound relates the *expected value / bias / mean squared error / variance* of the estimator $\hat{w}$ to the inverse of its Fisher information. An estimator that reaches the Cramér–Rao lower bound is said to be *consistent / efficient / unbiased*.

**Answer**

In order of the blanks:

- **unbiased** — $\mathbb{E}(\hat{w}) = w$ is the definition of unbiasedness.
- **mean squared error**, decomposed into **variance** + squared **bias**: $\mathrm{mse}(\hat{w}) = \mathrm{Var}(\hat{w}) + \mathrm{bias}^2(\hat{w})$.
- Let $\mathbb{E}(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. For $\hat{\mu}_n = \frac{1}{n+1}\sum_{i=1}^{n} X_i$, linearity of expectation gives

$$\mathbb{E}(\hat{\mu}_n) = \frac{1}{n+1}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{n}{n+1}\mu \neq \mu$$

  for finite $n$, so the estimator is biased. However,

$$\mathrm{bias}(\hat{\mu}_n) = -\frac{\mu}{n+1} \longrightarrow 0, \qquad \mathrm{Var}(\hat{\mu}_n) = \frac{n\sigma^2}{(n+1)^2} \longrightarrow 0.$$

  Thus it converges to $\mu$ and is **biased but consistent**.
- Taking $\hat{\mu} = X_1$ gives $\mathbb{E}(\hat{\mu}) = \mu$, so it is unbiased. But $\mathrm{Var}(\hat{\mu}) = \sigma^2$ for every sample size: the variance never approaches zero because all further observations are ignored. Therefore $X_1$ is **unbiased but not consistent**.
- $\prod_i p(x_i; w)$ is the **likelihood** function.
- $p(\cdot; w)$ is usually a **density**, so any finite set of points has probability measure zero.
- Cramér–Rao bounds the **variance**: $\mathrm{Var}(\hat{w}) \geq 1/\mathrm{Fisher}$; an estimator attaining the bound is **efficient**.

**Question 11: Unsupervised Machine Learning**

Which statements about Unsupervised Machine Learning are true? Multiple answers are possible.

- ☐ Underfitting means that a model is adapted to special training characteristics such as noisy measurements or outliers.
- ☐ Projection methods project data into a higher-dimensional space while keeping their neighborhoods.
- ☐ The model selection procedure is called “training” or “learning”.
- ☐ Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models.
- ☐ In parametric models, every parameter (vector) realization represents a model.
- ☐ $k$-nearest neighbors is a nonparametric model.
- ☐ Generative models do not aim at modeling the data creation process.
- ☐ Unsupervised Machine Learning algorithms learn patterns from unlabeled data.

**Answer**

- ✗ **A.** Adapting to noise and outliers of the training set is *over*fitting; underfitting means the model is too simple to capture the structure at all.
- ✗ **B.** Projection methods map into a *lower*-dimensional space.
- ✗ **C.** Training/learning fits the parameters of a model *within* a model class; model selection is the separate step of choosing among models.
- ✓ **D.** All of these work without labels — classic unsupervised toolbox.
- ✓ **E.** A parametric model class is a family indexed by the parameter vector; each realization picks one model.
- ✓ **F.** $k$-NN keeps the training data itself instead of a fixed-size parameter vector — nonparametric.
- ✗ **G.** The opposite: generative models explicitly model the data-generating process.
- ✓ **H.** That is the definition of unsupervised learning.

**Question 12: Expectation Maximization (EM) and Maximum Entropy — version A**

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct? Multiple answers are possible.

- ☐ The EM algorithm can be applied to hidden Markov models and mixture models.
- ☐ In the M-step, the model parameters are updated.
- ☐ The E-step increases the upper bound on the log-likelihood.
- ☐ In EM, one deals with problems that have hidden variables as well as model parameters.
- ☐ The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ Hidden state models in general have analytical solutions for the likelihood.
- ☐ In the E-step, the hidden variables parameters are updated.
- ☐ The Maximum Entropy approach requires minimal prior assumptions to assign a-priori probabilities.

**Answer**

- ✓ **A.** Baum–Welch (HMMs) and mixture-model fitting are *the* classic EM applications.
- ✓ **B.** The M-step maximizes the expected complete-data log-likelihood with respect to the parameters.
- ✗ **C.** EM works with a *lower* bound on the log-likelihood; there is no upper bound involved.
- ✓ **D.** Exactly the EM setting: latent variables plus model parameters.
- ✗ **E.** Discrete entropy is $\geq 0$ but *not* bounded by 1: a uniform distribution over $k$ outcomes has entropy $\ln k > 1$ for $k \geq 3$.
- ✗ **F.** Latent-variable likelihoods are generally analytically intractable — that is precisely why EM exists.
- ✓ **G.** The E-step (re-)estimates the hidden variables via their posterior given the current parameters.

✓ **H.** MaxEnt chooses the least-informative distribution consistent with the known constraints.

**Question 13: Expectation Maximization (EM) and Maximum Entropy — version B**

*Asked multiple times.*

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct? Multiple answers are possible.

- ☐ In the M-step, the hidden variables are updated.
- ☐ Maximum Entropy requires minimal prior assumptions to assign a-priori probabilities.
- ☐ The E-step increases the upper bound on the log-likelihood.
- ☐ In the E-step, the model parameters are updated.
- ☐ The EM algorithm can be applied to hidden Markov models and mixture models.
- ☐ The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ The E-step increases the lower bound on the log-likelihood.
- ☐ In EM, one deals with problems that have hidden variables as well as model parameters.

**Answer**

× **A.** Swapped: the M-step updates the *model parameters*.

✓ **B.** MaxEnt = minimal prior assumptions.

× **C.** Lower bound, not upper bound.

× **D.** Swapped: the E-step updates the estimates of the *hidden variables*.

✓ **E.** HMMs (Baum–Welch) and mixture models are the standard EM examples.

× **F.** Entropy can exceed 1 (uniform over $k \geq 3$ outcomes: $\ln k > 1$).

✓ **G.** The E-step tightens the lower bound so that it touches the log-likelihood at the current parameters.

✓ **H.** That is the EM setting.

## PCA, Kernel PCA, and ICA

**Question 14: Principal Component Analysis — statements, version A**

Which statements about Principal Component Analysis (PCA) are correct? Multiple answers are possible.

- ☐ The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- ☐ The singular values from the singular value decomposition of the data matrix $X$ are equal to the eigenvalues of the eigendecomposition of $X^\top X$.
- ☐ The PCA solution is unique, i.e. there is only one solution.
- ☐ Eigenvectors can be computed iteratively with Oja's rule.

- ☐ The sample variance of the $k$-th projection is equal to the $k$-th eigenvalue of the sample covariance matrix.
- ☐ The matrix of eigenvectors $U$ is an orthogonal matrix.
- ☐ The sample covariance matrix is symmetric and positive semidefinite.
- ☐ PCA transforms data into a new coordinate system such that the data has the largest mean value along the first coordinate, the second largest mean value along the second coordinate, and so on.

**Answer**

- ✓ **A.** Principal components are sorted by eigenvalue; the first one carries the largest.
- × **B.** Near-miss: the singular values are the *square roots* of the eigenvalues of $X^\top X$: $\sigma_i = \sqrt{\lambda_i}$.
- × **C.** Not unique — every eigenvector can be sign-flipped, and with equal eigenvalues the directions are not even determined.
- ✓ **D.** Oja's rule is the classic iterative (Hebbian) method to extract the leading eigenvector.
- ✓ **E.** The variance of the data projected onto $u_k$ is exactly $\lambda_k$.
- ✓ **F.** Eigenvectors of a symmetric matrix can be chosen orthonormal, so $U^\top U = I$.
- ✓ **G.** Any sample covariance matrix is symmetric and positive semidefinite by construction.
- × **H.** PCA orders directions by *variance*, not by mean value.

**Question 15: Principal Component Analysis — statements, version B**

We keep the notation of the lecture slides: $X$ is the data matrix and $U$ is the matrix of eigenvectors of $X^\top X$. Which statements about PCA are true? Multiple answers are possible.

- ☐ PCA transforms data into a new coordinate system such that the data has the largest variance along the first coordinate, the second largest variance along the second coordinate, and so on.
- ☐ The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- ☐ If there are $n$ data samples, each of which is $d$-dimensional, the matrix $U$ has dimension $n \times d$.
- ☐ The sample covariance matrix is orthogonal and positive semidefinite.
- ☐ The singular values from the singular value decomposition of the data matrix $X$ are equal to the square root of the eigenvalues of the eigendecomposition of $X^\top X$.
- ☐ The eigenvalues obtained from the eigendecomposition of $X^\top X$ are all positive and sum up to 1.
- ☐ PCA is commonly used to visualize data by downprojecting it to a smaller number of dimensions, keeping the most relevant information.
- ☐ The matrix $U$ is a symmetric matrix.

**Answer**

- ✓ **A.** That is the variance-maximization view of PCA.
- ✓ **B.** Correct, as in version A.
- × **C.** $U$ collects the eigenvectors of the $d \times d$ matrix $X^\top X$, so it is $d \times d$ — the sample count $n$ does not appear.
- × **D.** Symmetric and PSD, yes; *orthogonal*, no — a covariance matrix is generally not orthogonal.
- ✓ **E.** $\sigma_i = \sqrt{\lambda_i}$ — this is the correct version of the near-miss in version A.
- × **F.** Eigenvalues are only *non-negative* (zero is possible), and they sum to the total variance ($\mathrm{tr}(X^\top X)$), not to 1.
- ✓ **G.** Standard use case: project to 2–3 dimensions for visualization while keeping maximal variance.
- × **H.** $U$ is orthogonal ($U^\top U = I$), not symmetric in general.

## Question 16: Picture-based interpretation question: PCA, ICA, and EM

The original exam contained three picture-based subquestions. State which interpretation is more likely in each case.

1. PCA and ICA are applied to a 2D cloud of points, leading to outcomes A and B. Which picture is more likely the result of PCA and which one the result of ICA?
2. Two plots show a dataset in a coordinate system spanned by principal components. Which is more likely: picture 1 is shown in PC2 vs. PC1 and picture 2 in PC3 vs. PC1, or vice versa?
3. The left figure shows an original dataset, while figures $a$ and $b$ show results of running EM for a mixture of two Gaussians. Which is more likely: figure $a$ is after 5 iterations and figure $b$ after 20 iterations, or the other way around?

**Answer**

- **PCA vs. ICA:** PCA always returns *orthogonal* axes aligned with the directions of maximal variance, so the picture with perpendicular axes is PCA (A). ICA finds statistically independent directions, which are in general *not* orthogonal and follow the actual (possibly skewed) structure of the cloud (B).
- **PC order:** the projected variance decreases with the component index, so the scatter along PC2 must be larger than along PC3. The plot with the larger vertical spread is PC2 vs. PC1 (picture 1); the flatter one is PC3 vs. PC1 (picture 2).
- **EM iterations:** early in EM the two Gaussians still overlap and sit near their initialization; after more iterations they have converged onto the two clusters. Hence $a$ (poorly separated) is after 5 iterations and $b$ (well fitted) after 20.

## Question 17: PCA computation on a rank-1 matrix — projected second sample

*Asked multiple times.*

You are given four data samples, each of dimension 3. The data matrix reads

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}.$$

The eigenvectors of $X^\top X$ are

$$u_1 = \frac{1}{\sqrt{3}}(1,1,1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(-1,0,1)^\top, \qquad u_3 = \frac{1}{\sqrt{2}}(-1,1,0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix $Y$ and enter the components of the projected second feature vector $y_2$. Round to two decimal places.

**Answer**

Put the eigenvectors into the columns of $U = (u_1, u_2, u_3)$. The PCA coordinates are

$$Y = XU, \qquad Y_{ik} = x_i^\top u_k.$$

Thus every entry is the scalar projection of sample $x_i$ onto direction $u_k$. For the second sample $x_2 = (2,2,2)^\top$:

$$y_2^{(1)} = \frac{2+2+2}{\sqrt{3}} = \frac{6}{\sqrt{3}} = 2\sqrt{3} \approx \mathbf{3.46},$$

$$y_2^{(2)} = \frac{-2+0+2}{\sqrt{2}} = \mathbf{0}, \qquad y_2^{(3)} = \frac{-2+2+0}{\sqrt{2}} = \mathbf{0}.$$

Repeating the same three scalar products for all four rows gives the complete projection matrix:

$$Y = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 2\sqrt{3} & 0 & 0 \\ 3\sqrt{3} & 0 & 0 \\ 4\sqrt{3} & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} 1.73 & 0.00 & 0.00 \\ 3.46 & 0.00 & 0.00 \\ 5.20 & 0.00 & 0.00 \\ 6.93 & 0.00 & 0.00 \end{pmatrix}.$$

Every row of $X$ is a multiple of $(1,1,1)^\top$: the matrix has rank 1, so all variation lies along $u_1$ and the other two coordinates vanish for every sample. The requested row is therefore

$$y_2 = (\mathbf{3.46}, \mathbf{0.00}, \mathbf{0.00}).$$

**Question 18: PCA computation on a centered matrix — projected third sample**

*Asked multiple times.*

You are given four data samples, each of dimension 3. The data matrix reads

$$X = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix}.$$

The eigenvectors of the covariance matrix are

$$u_1 = \frac{1}{\sqrt{2}}(-1,0,1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(1,0,1)^\top, \qquad u_3 = (0,1,0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix $Y$ and enter the components of the projected third sample $y_3$. Round to two decimal places.

**Answer**

Again, $U = (u_1, u_2, u_3)$ and $Y = XU$, so $Y_{ik} = x_i^\top u_k$. For $x_3 = (-3, 0, 3)^\top$:

$$y_3^{(1)} = \frac{(-3)(-1) + 0 + 3 \cdot 1}{\sqrt{2}} = \frac{6}{\sqrt{2}} \approx \mathbf{4.24},$$

$$y_3^{(2)} = \frac{-3 + 0 + 3}{\sqrt{2}} = \mathbf{0}, \qquad y_3^{(3)} = 0 \cdot (-3) + 1 \cdot 0 + 0 \cdot 3 = \mathbf{0}.$$

For a general row $x_i = (-i, 0, i)^\top$, these products are $x_i^\top u_1 = i\sqrt{2}$ and $x_i^\top u_2 = x_i^\top u_3 = 0$. Hence the complete matrix is

$$Y = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 2\sqrt{2} & 0 & 0 \\ 3\sqrt{2} & 0 & 0 \\ 4\sqrt{2} & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} 1.41 & 0.00 & 0.00 \\ 2.83 & 0.00 & 0.00 \\ 4.24 & 0.00 & 0.00 \\ 5.66 & 0.00 & 0.00 \end{pmatrix}.$$

Thus $y_3 = (\mathbf{4.24}, \mathbf{0.00}, \mathbf{0.00})$. All rows are multiples of $(-1, 0, 1)^\top$, so the data is one-dimensional along $u_1$.

**Question 19: PCA computation with a repeated sample — projected third sample**

You are given four data samples, each of dimension 3, collected in the data matrix

$$X = \begin{pmatrix} -4 & 0 & -4 \\ -2 & 0 & 2 \\ 3 & 0 & 3 \\ 3 & 0 & 3 \end{pmatrix}.$$

Further assume that the eigenvectors of the covariance matrix of this data are

$$u_1 = \frac{1}{\sqrt{2}}(1, 0, 1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)^\top, \qquad u_3 = (0, 1, 0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix $Y$ and enter the components of the projected third sample $y_3$. Round to two decimal places.

**Answer**

Using the projection convention in the question, put the stated directions into $U = (u_1, u_2, u_3)$ and calculate $Y = XU$. For $x_3 = (3, 0, 3)^\top$:

$$y_3^{(1)} = \frac{3 \cdot 1 + 0 + 3 \cdot 1}{\sqrt{2}} = \frac{6}{\sqrt{2}} \approx \mathbf{4.24}, \qquad y_3^{(2)} = \frac{3 \cdot (-1) + 0 + 3 \cdot 1}{\sqrt{2}} = \mathbf{0}, \qquad y_3^{(3)} = \mathbf{0}.$$

The other rows are computed in exactly the same way:

$$Y = \begin{pmatrix} -4\sqrt{2} & 0 & 0 \\ 0 & 2\sqrt{2} & 0 \\ 3\sqrt{2} & 0 & 0 \\ 3\sqrt{2} & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} -5.66 & 0.00 & 0.00 \\ 0.00 & 2.83 & 0.00 \\ 4.24 & 0.00 & 0.00 \\ 4.24 & 0.00 & 0.00 \end{pmatrix}.$$

The third sample lies exactly on the $(1, 0, 1)$ direction, so only its first component survives:

$$y_3 = (\mathbf{4.24}, \mathbf{0.00}, \mathbf{0.00}).$$

The fourth sample is identical to the third, so it projects to the same point.

**Question 20: Kernel PCA**

Which statements about Kernel PCA are correct? Use the notation from the lecture: data points $x_1, \ldots, x_n \in \mathcal{X} = \mathbb{R}^m$, kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, feature map $\phi : \mathcal{X} \to \mathcal{H}$, feature covariance matrix

$$C = \frac{1}{n} \sum_{i=1}^{n} \phi(x_i)\phi(x_i)^\top,$$

and Gram matrix $K$ with entries $K_{ij} = \phi(x_i)^\top \phi(x_j)$. Multiple answers are possible.

- ☐ A disadvantage of Kernel PCA is that $\phi$ always needs to be computed explicitly, which is computationally costly.
- ☐ In general, any nonnegative and symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defines a kernel.
- ☐ To compute the projection of a data point $x$ onto the eigenvectors of $K$, you need to center $x$ and $K$, as well as to normalize the eigenvectors of $K$.
- ☐ The main task of Kernel PCA is to solve the eigenvalue problem of $C$ in feature space. This can be achieved by solving the eigenvalue problem of $K$ in a Euclidean space.
- ☐ In general, $\phi$ is a nonlinear map that maps into a higher-dimensional feature space $\mathcal{H}$.
- ☐ Linear PCA can be seen as a special case of Kernel PCA by using $k(x, y) = x^\top y$ as the associated kernel function.
- ☐ For Kernel PCA, it is essential to center $K$ before computing its eigendecomposition, since the eigenvalues of $K$ can in general be negative, which leads to inconclusive results.
- ☐ $K$ and $C$ always have the same dimensionality and the same rank.

**Answer**

- ✗ **A.** The kernel trick exists precisely so that $\phi$ is *never* computed explicitly — only inner products via $k$.
- ✗ **B.** A kernel must be symmetric *and positive semidefinite* (Mercer); nonnegativity of the function values is neither necessary nor sufficient.
- ✓ **C.** Centering (in feature space, applied to both $K$ and the new point) and normalizing the eigenvectors are both required for correct projections.
- ✓ **D.** The eigenproblem of $C$ in $\mathcal{H}$ reduces to the $n \times n$ eigenproblem of the Gram matrix $K$.
- ✓ **E.** Typical feature maps are nonlinear and map into a higher- (possibly infinite-)

dimensional space.

✓ **F.** With the linear kernel $k(x, y) = x^\top y$, Kernel PCA reproduces ordinary PCA.

× **G.** Tricky: centering *is* needed, but the reason is wrong — $K$ is PSD, so its eigenvalues are never negative. Centering removes the feature-space mean.

× **H.** They share the same *nonzero* eigenvalues (hence rank), but $K$ is $n \times n$ while $C$ lives in feature space — the dimensionalities differ in general.

## Question 21: Independent Component Analysis — version A

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

☐ Maximizing the negentropy maximizes the mutual information.

☐ In general, ICA components are neither ranked nor orthogonal.

☐ Non-Gaussianity can be measured by the second cumulant.

☐ Criteria for independence are mutual information between components and non-Gaussianity of components.

☐ The ICA solution is unique.

☐ For ICA, one needs at least as many sources as observations.

☐ Infomax maximizes the mutual information between components.

☐ FastICA relies on whitening and rotating the data.

## Answer

× **A.** Maximizing negentropy (non-Gaussianity) *minimizes* the mutual information between components.

✓ **B.** Unlike PCA, the components carry no natural order and need not be orthogonal.

× **C.** The second cumulant is the *variance* — it is blind to non-Gaussianity; one uses the 3rd/4th cumulants (skewness, kurtosis).

✓ **D.** Those are exactly the two standard independence criteria.

× **E.** Only unique up to permutation, sign, and scaling of the components.

× **F.** Reversed: one needs at least as many *observations* as sources.

× **G.** Infomax maximizes the output entropy and thereby *minimizes* the mutual information between components.

✓ **H.** FastICA: whiten the data first, then rotate to maximize non-Gaussianity.

## Question 22: Independent Component Analysis — version B

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

☐ Independence can be quantified by the mutual information between components or the non-Gaussianity of components.

☐ Fast ICA uses whitening and rotating the data.

- ☐ Maximizing the negentropy minimizes the mutual information.
- ☐ For non-Gaussian sources, the ICA solution is unique up to permutation and scaling.
- ☐ Non-Gaussianity can be measured by the second and third cumulant.
- ☐ For ICA, the number of sources must be smaller than, or equal to, the number of observations.
- ☐ ICA components are ranked and orthogonal.
- ☐ Infomax maximizes the mutual information between components.

**Answer**

- ✓ **A.** Mutual information and non-Gaussianity are the standard independence criteria.
- ✓ **B.** Whitening + rotation is the FastICA recipe.
- ✓ **C.** This is the correct direction: negentropy up ⇔ mutual information down.
- ✓ **D.** Identifiability result for non-Gaussian sources: unique up to permutation and scaling (incl. sign).
- × **E.** Near-miss: the *third* cumulant (skewness) does measure non-Gaussianity, but the *second* is just the variance — including it makes the statement false.
- ✓ **F.** #sources ≤ #observations is the standard requirement.
- × **G.** Neither ranked nor orthogonal in general.
- × **H.** Infomax *minimizes* the mutual information between components.

## Question 23: Independent Component Analysis — version C

*Asked multiple times.*

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

- ☐ For ICA, one needs more sources than observations.
- ☐ Maximizing the negentropy maximizes the mutual information.
- ☐ Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☐ The ICA solution is unique.
- ☐ Fast ICA relies on whitening and rotating the data.
- ☐ Infomax maximizes the mutual information between components.
- ☐ Non-Gaussianity can be measured by the second cumulant.
- ☐ In general, ICA components are neither ranked nor orthogonal.

**Answer**

- × **A.** Reversed: at least as many *observations* as sources.
- × **B.** Maximizing negentropy *minimizes* mutual information.
- ✓ **C.** The two standard independence criteria.

× **D.** Unique only up to permutation, sign, and scaling.

✓ **E.** Whitening + rotation is the FastICA scheme.

× **F.** Infomax minimizes the mutual information between components.

× **G.** The second cumulant is the variance — no information about Gaussianity.

✓ **H.** Correct — no ranking, no orthogonality.

## Factor Analysis

**Question 24: Factor Analysis: low-rank covariance part**

In factor analysis, part of the covariance function is formed by

$$C_U = UU^\top = \sum_{i=1}^{\ell} u_i u_i^\top,$$

where $U \in \mathbb{R}^{m\times\ell}$, the vectors $u_i$ are the columns of $U$, and $\ell < m$. Which constraints does this impose on $C_U$? Multiple answers are possible.

☐ The maximum rank of the matrix is $\ell$.

☐ The number of zero eigenvalues of $C_U$ is greater or equal to $m - \ell$.

☐ Any covariance matrix in $m$ dimensions can be written in the form above.

☐ $C_U$ is singular.

**Answer**

✓ **A.** $C_U$ is a sum of $\ell$ rank-1 terms, so $\operatorname{rank}(C_U) \leq \ell$.

✓ **B.** A symmetric $m \times m$ matrix of rank $\leq \ell$ has at least $m - \ell$ zero eigenvalues.

× **C.** A full-rank covariance matrix (rank $m$) cannot be written with only $\ell < m$ columns — that is exactly why FA adds the diagonal noise term $\Psi$.

✓ **D.** Rank $< m$ implies $\det C_U = 0$, i.e. $C_U$ is singular.

**Question 25: Factor Analysis: dimensions of the objects**

We keep the notation from the lecture. Here $n$ is the number of data samples, $m$ is the dimension of each sample, $\ell$ is the number of factors, $y$ is the factor vector, $U$ is the factor loading matrix, $\Psi$ is the diagonal noise covariance matrix, and $\varepsilon$ is the noise vector. Which statements are correct? Multiple answers are possible.

☐ $\Psi$ has shape $m \times m$.

☐ $\Psi$ has shape $\ell \times \ell$.

☐ $\Psi$ has shape $n \times \ell$.

☐ $U$ has shape $m \times \ell$.

☐ $U$ has shape $m \times n$.

☐ $U$ has shape $\ell \times m$.

- ☐ $\varepsilon$ has dimension $m$.
- ☐ $\varepsilon$ has dimension $\ell$.
- ☐ $\varepsilon$ has dimension $n$.
- ☐ $y$ has dimension $m$.
- ☐ $y$ has dimension $\ell$.
- ☐ $y$ has dimension $n$.

**Answer**

Everything follows from the FA model equation

$$x = Uy + \varepsilon, \qquad x \in \mathbb{R}^m,$$

read off left to right: to produce an $m$-dimensional observation from an $\ell$-dimensional factor vector, $U$ must be $m \times \ell$ (**D**); the factor vector $y$ has dimension $\ell$ (**K**); the noise $\varepsilon$ is added to the observation, so it has dimension $m$ (**G**); and its covariance $\Psi = \mathrm{Cov}(\varepsilon)$ is $m \times m$ (**A**). The number of samples $n$ never appears in any of these shapes — every option containing $n$ is automatically wrong.

**Question 26: Factor Analysis: compute the first row of the covariance matrix**

You are given the factor loading matrix

$$U = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

and the diagonal noise covariance matrix

$$\Psi = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Compute the first row of the covariance matrix

$$C = UU^\top + \Psi.$$

**Answer**

In general, entry $(i, j)$ of $UU^\top$ is the dot product of row $i$ and row $j$ of $U$:

$$(UU^\top)_{ij} = \sum_{k=1}^{\ell} U_{ik} U_{jk}.$$

For the first row, take $(1, 2)$ and form its dot product with each row of $U$:

$$(UU^\top)_{11} = 1 \cdot 1 + 2 \cdot 2 = 5, \qquad (UU^\top)_{12} = 1 \cdot 3 + 2 \cdot 4 = 11, \qquad (UU^\top)_{13} = 1 \cdot 5 + 2 \cdot 6 = 17.$$

Hence the first row of $UU^\top$ is $(5, 11, 17)$. The first row of $\Psi$ is $(5, 0, 0)$, so

$$C_{1,:} = (5, 11, 17) + (5, 0, 0) = (\mathbf{10}, \mathbf{11}, \mathbf{17}).$$

The diagonal noise term changes $C_{11}$ but not the off-diagonal entries in this row.

**Question 27: Factor Analysis: conceptual statements**

Which statements about Factor Analysis (FA) are correct? Multiple answers are possible.

- ☐ The factors are unique only up to orthogonal transformations (rotations).
- ☐ Correlations between observations can only be explained by factors since the noise covariance matrix $\Psi$ is diagonal.
- ☐ FA describes the variability of observations in terms of latent variables and noise.
- ☐ The factor loading matrix $U$ describes the independent part of the covariance of the observations.
- ☐ FA decomposes the covariance matrix of the data as $C = UU^\top + \Psi$.
- ☐ FA assumes Gaussian noise and non-Gaussian latent variables.
- ☐ Factors explain the variance of the observations, while the correlation is explained by noise.

**Answer**

- ✓ **A.** For any rotation $R$: $(UR)(UR)^\top = UU^\top$, so $U$ is only determined up to rotations.
- ✓ **B.** With diagonal $\Psi$, all off-diagonal covariance must come from $UU^\top$ — i.e. from the factors.
- ✓ **C.** That is the FA model: shared latent factors plus independent noise.
- × **D.** Swapped: $UU^\top$ is the *shared/correlated* part; the independent part is $\Psi$.
- ✓ **E.** Exactly the FA covariance decomposition.
- × **F.** Standard FA assumes *both* Gaussian factors and Gaussian noise (that is what makes it a tractable linear-Gaussian model; ICA is the non-Gaussian relative).
- × **G.** Reversed: factors explain the *correlations*, noise the independent leftover variance.

## Clustering, Mixtures, and Projection Methods

**Question 28: $k$-means: one iteration, dataset A**

Assume you want to do $k$-means on the following 2-dimensional dataset:

$$X = \{(1,0), (0,1), (-1,0), (0,-1), (2,0), (2,3), (5,0)\}.$$

The initial cluster centers are $\mu = (0,1)$ and $\nu = (2,2)$. Compute the coordinates of the new cluster centers $\mu^{\text{new}}$ and $\nu^{\text{new}}$ after one iteration of $k$-means.

**Answer**

**Assignment step.** For $x = (x_1, x_2)$ and center $c = (c_1, c_2)$, use

$$d^2(x, c) = (x_1 - c_1)^2 + (x_2 - c_2)^2.$$

The square root is unnecessary because the smaller squared distance also has the smaller

ordinary distance. With $\mu = (0, 1)$ and $\nu = (2, 2)$:

| $x$ | $d^2(x, \mu)$ | $d^2(x, \nu)$ | assignment |
|---|---|---|---|
| $(1, 0)$ | 2 | 5 | $\mu$ |
| $(0, 1)$ | 0 | 5 | $\mu$ |
| $(-1, 0)$ | 2 | 13 | $\mu$ |
| $(0, -1)$ | 4 | 13 | $\mu$ |
| $(2, 0)$ | 5 | 4 | $\nu$ |
| $(2, 3)$ | 8 | 1 | $\nu$ |
| $(5, 0)$ | 26 | 13 | $\nu$ |

**Update step.** A new center is the component-wise mean of all points assigned to that cluster:

$$\begin{aligned}\mu^{\text{new}} &= \tfrac{1}{4}\big[(1,0) + (0,1) + (-1,0) + (0,-1)\big] \\ &= \left(\frac{1+0-1+0}{4}, \frac{0+1+0-1}{4}\right) = (\mathbf{0}, \mathbf{0}), \\ \nu^{\text{new}} &= \tfrac{1}{3}\big[(2,0) + (2,3) + (5,0)\big] \\ &= \left(\frac{2+2+5}{3}, \frac{0+3+0}{3}\right) = (\mathbf{3}, \mathbf{1}).\end{aligned}$$

## Question 29: $k$-means and affinity propagation

Which statements about $k$-means and affinity propagation are correct? Multiple answers are possible.

- ☐ The $k$-means objective is to minimize the within-cluster sum of squared deviations.
- ☐ In affinity propagation: the larger the preference, the smaller is the number of cluster centers.
- ☐ In affinity propagation: the larger the damping factor, the faster the convergence.
- ☐ In affinity propagation, cluster centers are data points.
- ☐ In affinity propagation, the similarity between two points is typically chosen as their negative squared distance.
- ☐ $k$-means always converges to the global optimum.
- ☐ The parameter $k$ in $k$-means is automatically chosen by the algorithm to best fulfill the optimization objective.
- ☐ $k$-means is sensitive to initialization of the cluster centers.

## Answer

- ✓ **A.** $k$-means minimizes $\sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$.
- × **B.** Reversed: the preference is a point's a-priori suitability as an exemplar — the larger it is, the *more* cluster centers emerge.
- × **C.** Damping stabilizes the message updates against oscillations; a larger damping factor makes convergence *slower*, not faster.
- ✓ **D.** Cluster centers (exemplars) are always actual data points.
- ✓ **E.** $s(i, k) = -\|x_i - x_k\|^2$ is the standard choice.

× **F.** $k$-means only reaches a *local* optimum of its objective.

× **G.** $k$ is a user-chosen hyperparameter — the algorithm never selects it.

✓ **H.** Different initial centers can lead to different local optima; that is why one restarts $k$-means several times.

### Question 30: $k$-means: one iteration, dataset B

Assume you want to do $k$-means on the following 2-dimensional dataset:

$$X = \{(1,0), (2,-1), (1,-2), (0,-1), (2,0), (2,1), (3,0), (3,1)\}.$$

The initial cluster centers are $\mu = (0,-2)$ and $\nu = (4,1)$. Compute the coordinates of the new cluster centers $\mu^{\text{new}}$ and $\nu^{\text{new}}$ after one iteration of $k$-means.

### Answer

**Assignment step.** Compare $d^2(x,c) = (x_1 - c_1)^2 + (x_2 - c_2)^2$ for the two initial centers:

| $x$ | $d^2(x,\mu)$ | $d^2(x,\nu)$ | assignment |
|---|---|---|---|
| $(1,0)$ | 5 | 10 | $\mu$ |
| $(2,-1)$ | 5 | 8 | $\mu$ |
| $(1,-2)$ | 1 | 18 | $\mu$ |
| $(0,-1)$ | 1 | 20 | $\mu$ |
| $(2,0)$ | 8 | 5 | $\nu$ |
| $(2,1)$ | 13 | 4 | $\nu$ |
| $(3,0)$ | 13 | 2 | $\nu$ |
| $(3,1)$ | 18 | 1 | $\nu$ |

**Update step.** Take the component-wise mean in each cluster:

$$\begin{aligned}\mu^{\text{new}} &= \tfrac{1}{4}\big[(1,0) + (2,-1) + (1,-2) + (0,-1)\big] \\ &= \left(\frac{1+2+1+0}{4}, \frac{0-1-2-1}{4}\right) = (\mathbf{1}, \mathbf{-1}),\end{aligned}$$

$$\begin{aligned}\nu^{\text{new}} &= \tfrac{1}{4}\big[(2,0) + (2,1) + (3,0) + (3,1)\big] \\ &= \left(\frac{2+2+3+3}{4}, \frac{0+1+0+1}{4}\right) = (\mathbf{2.5}, \mathbf{0.5}).\end{aligned}$$

### Question 31: $k$-means: one iteration, dataset C

Assume you want to do $k$-means on the following 2-dimensional dataset:

$$X = \{(1,-1), (0,0), (-1,-1), (0,-2), (2,0), (2,3), (5,0)\}.$$

The initial cluster centers are $\mu = (0,0)$ and $\nu = (2,1)$. Compute the coordinates of the new cluster centers $\mu^{\text{new}}$ and $\nu^{\text{new}}$ after one iteration of $k$-means.

**Answer**

**Assignment step.** Compare the squared Euclidean distances to $\mu = (0, 0)$ and $\nu = (2, 1)$:

| $x$ | $d^2(x, \mu)$ | $d^2(x, \nu)$ | assignment |
|---|---|---|---|
| $(1, -1)$ | 2 | 5 | $\mu$ |
| $(0, 0)$ | 0 | 5 | $\mu$ |
| $(-1, -1)$ | 2 | 13 | $\mu$ |
| $(0, -2)$ | 4 | 13 | $\mu$ |
| $(2, 0)$ | 4 | 1 | $\nu$ |
| $(2, 3)$ | 13 | 4 | $\nu$ |
| $(5, 0)$ | 25 | 10 | $\nu$ |

**Update step.** Average the coordinates of the assigned points:

$$\begin{aligned}\mu^{\text{new}} &= \tfrac{1}{4}[(1,-1)+(0,0)+(-1,-1)+(0,-2)] \\ &= \left(\frac{1+0-1+0}{4}, \frac{-1+0-1-2}{4}\right) = (\mathbf{0}, \mathbf{-1}),\end{aligned}$$

$$\begin{aligned}\nu^{\text{new}} &= \tfrac{1}{3}[(2,0)+(2,3)+(5,0)] \\ &= \left(\frac{2+2+5}{3}, \frac{0+3+0}{3}\right) = (\mathbf{3}, \mathbf{1}).\end{aligned}$$

**Question 32: $k$-means: one iteration, dataset D**

Assume you want to do $k$-means on the following 2-dimensional dataset:

$$X = \{(0,1), (1,2), (2,1), (1,0), (0,2), (-1,2), (0,3), (-1,3)\}.$$

The initial cluster centers are $\mu = (2, 0)$ and $\nu = (-1, 4)$. Compute the coordinates of the new cluster centers $\mu^{\text{new}}$ and $\nu^{\text{new}}$ after one iteration of $k$-means. A sketch of the data points helps: the assignment step can largely be read off graphically. Give the values with one decimal place.

**Answer**

**Assignment step.** Use $d^2(x, c) = (x_1 - c_1)^2 + (x_2 - c_2)^2$ and compare the squared distances to $\mu = (2, 0)$ and $\nu = (-1, 4)$:

| $x$ | $d^2_\mu$ | $d^2_\nu$ | $\to$ |
|---|---|---|---|
| $(0, 1)$ | 5 | 10 | $\mu$ |
| $(1, 2)$ | 5 | 8 | $\mu$ |
| $(2, 1)$ | 1 | 18 | $\mu$ |
| $(1, 0)$ | 1 | 20 | $\mu$ |
| $(0, 2)$ | 8 | 5 | $\nu$ |
| $(-1, 2)$ | 13 | 4 | $\nu$ |
| $(0, 3)$ | 13 | 2 | $\nu$ |
| $(-1, 3)$ | 18 | 1 | $\nu$ |

**Update step.** Take the component-wise mean in each cluster:

$$\begin{aligned}\mu^{\text{new}} &= \tfrac{1}{4}[(0,1)+(1,2)+(2,1)+(1,0)] \\ &= \left(\frac{0+1+2+1}{4}, \frac{1+2+1+0}{4}\right) = (\mathbf{1.0}, \mathbf{1.0}),\end{aligned}$$

$$\begin{aligned}\nu^{\text{new}} &= \tfrac{1}{4}[(0,2)+(-1,2)+(0,3)+(-1,3)] \\ &= \left(\frac{0-1+0-1}{4}, \frac{2+2+3+3}{4}\right) = (-\mathbf{0.5}, \mathbf{2.5}).\end{aligned}$$

**Question 33: $t$-SNE**

Which statements about $t$-distributed Stochastic Neighborhood Embedding are correct? Multiple answers are possible.

- ☐ It models the neighborhood in both observation and embedding space by a Gaussian distribution.
- ☐ It minimizes the Kullback–Leibler divergence between distributions in observation space and embedding space.
- ☐ It models the neighborhood of a data point in observation space proportional to a Student's $t$-distribution.
- ☐ It is usually trained by gradient descent.

**Answer**

- × **A.** Gaussian in the *observation* space only; the embedding space uses a Student's $t$-distribution (that is the "$t$" in $t$-SNE).
- ✓ **B.** The objective is $D_{\mathrm{KL}}(P \,\|\, Q)$ between the input-space and embedding-space similarity distributions.
- × **C.** Swapped: the $t$-distribution is used in the *embedding* space; the observation space uses Gaussians.
- ✓ **D.** The KL objective is minimized by (stochastic) gradient descent.

**Question 34: Scaling and Projection Methods — version A**

Which statements about Scaling and Projection Methods are correct? Multiple answers are possible.

- ☐ Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space.
- ☐ The objective in $t$-SNE is minimization of the KL divergence between the similarities of the high-dimensional inputs and the low-dimensional mapped outputs.
- ☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ Multidimensional scaling (MDS) keeps the distances between data points as well as possible.
- ☐ LLE in general performs linear mappings to realize neighborhood-preserving embeddings.
- ☐ In $t$-SNE, the similarity between data points $i$ and $j$ is given by their joint probability.

**Answer**

- ✓ **A.** Isomap is classical MDS on geodesic distances; the leading eigenvectors of the (centered) distance matrix give the embedding coordinates.
- ✓ **B.** That is exactly the $t$-SNE objective.
- ✓ **C.** LLE reconstructs every point as a weighted linear combination of its neighbors and preserves those weights in the embedding.
- × **D.** Scaling/projection methods go to a *lower*-dimensional space.
- × **E.** The Kohonen SOM update rules are *not* the gradient of any global energy function — there is no energy being minimized.
- ✓ **F.** Distance preservation is the definition of MDS.
- × **G.** LLE is a *nonlinear* embedding method — only the local reconstructions are linear.
- ✓ **H.** $t$-SNE uses symmetrized joint probabilities $p_{ij}$ and $q_{ij}$.

**Question 35: Projection and scaling methods — version B**

Which statements about projection and scaling methods are correct? Multiple answers are possible.

- ☐ Non-negative Matrix Factorization (NMF) performs a multiplicative decomposition of a positive semidefinite matrix into two positive semidefinite matrices.
- ☐ Locally Linear Embedding (LLE) is a linear embedding method that aims to preserve the neighborhood of each data point.
- ☐ Multidimensional Scaling (MDS) aims to find a low-dimensional representation of the data that preserves pairwise Euclidean distances.
- ☐ Stochastic Neighbor Embedding (SNE) defines the similarity of two data points as a conditional probability while $t$-SNE defines it as a joint probability.
- ☐ Isomap measures distances between data points as the length of the shortest path along a neighborhood graph.

**Answer**

- × **A.** NMF factorizes an *entrywise non-negative* matrix into entrywise non-negative factors — positive semidefiniteness is a different concept and plays no role.
- × **B.** LLE preserves neighborhoods, but it is a *nonlinear* method.
- ✓ **C.** That is the definition of (metric) MDS.
- ✓ **D.** SNE uses conditionals $p_{j|i}$; $t$-SNE symmetrizes them into joints $p_{ij}$.
- ✓ **E.** Geodesic distance $\approx$ shortest path through the $k$-nearest-neighbor graph.

**Question 36: Scaling and projection methods — version C**

Tick the correct statements about scaling and projection methods. We collect the original data points in a matrix $X$ and the mapped points in a matrix $Y$. Multiple answers are possible.

- ☐ The $t$-distribution has heavier tails than the Gaussian distribution; thus in $t$-SNE the joint distributions $q_{ij}$ between $y_i$ and $y_j$ are computed using a $t$-distribution in order to

alleviate the crowding problem.

- ☐ Isomap tries to approximate the geodesic distance between two points on a manifold by constructing a neighborhood graph.
- ☐ In SNE and $t$-SNE, the perplexity term is computed using the entropy of the similarities $q_{ij}$ of the mapped data points $y_i$ and $y_j$.
- ☐ In SNE and $t$-SNE, the objective is the mean squared distance between $x_i$ and $y_i$.
- ☐ Locally linear embedding and multidimensional scaling are probabilistic methods that rely on minimizing the KL divergence between the distributions of $X$ and $Y$.
- ☐ In SNE and $t$-SNE, the computation of the similarity $p_{ij}$ between $x_i$ and $x_j$ relies on assuming a Gaussian distribution centered at $x_i$.

**Answer**

- ✓ **A.** The heavy tails give moderately distant pairs more room in the embedding — exactly the fix for the crowding problem.
- ✓ **B.** Isomap approximates geodesics by shortest paths in the neighborhood graph.
- × **C.** Tricky: perplexity is set on the *input-space* similarities $p_{j|i}$ (to choose the Gaussian bandwidths), not on the $q_{ij}$ of the mapped points.
- × **D.** The objective is the KL divergence between similarity distributions, not a mean squared distance ($x_i$ and $y_i$ don't even live in the same space).
- × **E.** Neither LLE nor MDS is probabilistic or KL-based — that describes SNE/$t$-SNE.
- ✓ **F.** $p_{j|i}$ comes from a Gaussian kernel centered at $x_i$.

## Question 37: Scaling and Projection Methods — version D

*Asked multiple times.*

Which statements about Scaling and Projection Methods are correct? Multiple answers are possible.

- ☐ Multidimensional scaling (MDS) keeps the distances between data points as well as possible.
- ☐ The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space.
- ☐ In $t$-SNE, the similarity between data points $i$ and $j$ is given by their joint probability.
- ☐ In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ The objective in $t$-SNE is minimization of the KL divergence between the similarities of the high-dimensional inputs and the low-dimensional mapped outputs.
- ☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ LLE in general performs linear mappings to realize neighborhood-preserving embeddings.

**Answer**

✓ **A.** Distance preservation is the definition of MDS.

× **B.** *Lower*-dimensional, not higher.

✓ **C.** Isomap = MDS on the geodesic distance matrix; its eigenvectors give the coordinates.

✓ **D.** $t$-SNE uses joint probabilities $p_{ij}$, $q_{ij}$.

× **E.** SOMs have no global energy function.

✓ **F.** That is the $t$-SNE objective.

✓ **G.** LLE's local reconstruction property.

× **H.** LLE is nonlinear overall — only the local reconstructions are linear.

**Question 38: Scaling and Projection Methods — version E**

Decide which statements about Scaling and Projection Methods are correct. Multiple answers are possible.

☐ The objective of scaling and projection methods is to project data into a lower-dimensional space, e.g. to allow better model selection.

☐ In self-organizing maps the objective is to minimize an energy (error) function.

☐ Projection pursuit performs projections such that the extracted signal is as Gaussian as possible.

☐ Generative Topographic Mapping (GTM) is a non-linear latent variable model which maps latent variables to observations.

☐ Isomap uses a geodesic distance matrix, whose largest eigenvectors are the coordinates of the projected space.

☐ The objective in $t$-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs.

☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.

☐ Multidimensional Scaling (MDS) projects data to a higher-dimensional space in order to increase the distance between data points.

**Answer**

✓ **A.** This is the correct version of the recurring distractor: the projection goes into a *lower*-dimensional space.

× **B.** The Kohonen SOM update rules are not the gradient of any energy function — nothing is being minimized.

× **C.** Reversed: projection pursuit looks for maximally *non*-Gaussian ("interesting") projections — a Gaussian projection is the least informative one.

✓ **D.** GTM maps a grid of latent points through a nonlinear function into data space — a nonlinear latent variable model.

✓ **E.** The eigenvectors belonging to the largest eigenvalues of the (centered) geodesic distance matrix give the Isomap coordinates.

✓ **F.** The $t$-SNE objective.

✓ **G.** LLE's defining property.

× **H.** MDS embeds into a *lower*-dimensional space while *preserving* distances — it neither adds dimensions nor inflates distances.

### Question 39: Non-negative Matrix Factorization (NMF)

As in the lecture, let

$$X = YU^\top, \qquad Y \in \mathbb{R}^{n\times\ell}, \qquad U \in \mathbb{R}^{m\times\ell},$$

with non-negativity constraints. Which statements about NMF are correct? Multiple answers are possible.

☐ NMF objectives are the KL divergence and the Euclidean distance.

☐ NMF is always exactly solvable.

☐ $X \in \mathbb{R}^{n\times m}$.

☐ The non-negativity constraints make the representation of the observations purely additive.

☐ The outer product representation reads $X = \sum_{k=1}^{\ell} y_k u_k^\top$.

☐ NMF factorizes any positive or negative definite matrix into two positive definite matrices.

### Answer

✓ **A.** The two standard NMF objectives are the squared Euclidean distance and the (generalized) KL divergence.

× **B.** Exact NMF is NP-hard in general; the multiplicative update rules only find local optima of the chosen objective.

✓ **C.** $Y \in \mathbb{R}^{n\times\ell}$ times $U^\top \in \mathbb{R}^{\ell\times m}$ gives $X \in \mathbb{R}^{n\times m}$.

✓ **D.** Since no negative entries exist, components can never cancel each other — the representation is purely additive (parts-based).

✓ **E.** Column-times-row expansion of the product: $X = \sum_k y_k u_k^\top$.

× **F.** Nonsense — definiteness plays no role in NMF, and a negative definite matrix cannot even have a non-negative factorization.

### Question 40: Mixturetown bread question

In Mixturetown there are two bakers, Albert ($A$) and Bibi ($B$). Their loaf weights are modeled as Gaussian distributions

$$f(x;\mu_A,\sigma_A), \qquad f(x;\mu_B,\sigma_B),$$

with

$$\mu_A = 1 \text{ kg}, \quad \sigma_A = 0.1 \text{ kg}, \qquad \mu_B = 1.5 \text{ kg}, \quad \sigma_B = 0.2 \text{ kg}.$$

Seventy-five percent of all loaves are made by Bibi. You observe a loaf of weight 1.1 kg. Determine the probability that Bibi made this loaf. Use the given density values

$$f(1.1;1,0.1) = 2.41971, \qquad f(1.1;1.5,0.2) = 0.26995.$$

Round to two decimal places.

**Answer**

The statement "75% are made by Bibi" supplies the priors $P(B) = 0.75$ and $P(A) = 1 - 0.75 = 0.25$. For a continuous observation, the values $f(x \mid A)$ and $f(x \mid B)$ are densities; Bayes' rule combines each density with its prior and then normalizes:

$$P(B \mid x = 1.1) = \frac{P(B)\, f(1.1; \mu_B, \sigma_B)}{P(A)\, f(1.1; \mu_A, \sigma_A) + P(B)\, f(1.1; \mu_B, \sigma_B)}$$

$$= \frac{0.75 \cdot 0.26995}{0.25 \cdot 2.41971 + 0.75 \cdot 0.26995} = \frac{0.20246}{0.60493 + 0.20246} = \frac{0.20246}{0.80739} \approx \mathbf{0.25}.$$

Despite Bibi's large prior, the loaf weight is much more typical of Albert's ovens.

## Question 41: Mixture model posterior probability

*Asked multiple times.*

Assume the following case:

- The prior probability of component $j$ is 0.1.
- The probability for observing a data sample $x$ given the total parameter set $\theta$ is 0.2.
- The likelihood for observing a data sample $x$, given component $j$ and corresponding parameter vector, is 0.5.

Compute the posterior probability of component $j$. Round to two decimal places.

**Answer**

Bayes' rule for a mixture component is

$$P(j \mid x) = \frac{\overbrace{P(j)}^{\text{prior}} \; \overbrace{p(x \mid j)}^{\text{component likelihood}}}{\underbrace{p(x \mid \theta)}_{\text{total evidence}}}.$$

The joint contribution of component $j$ and observation $x$ is $P(j)p(x \mid j) = 0.1 \cdot 0.5 = 0.05$. Normalizing by the total probability of observing $x$ gives

$$P(j \mid x) = \frac{0.05}{0.2} = \mathbf{0.25}.$$

## Question 42: Mixture model: infer the likelihood from posterior, prior, and evidence

*Asked multiple times.*

Assume the following case:

- The prior probability of component $j$ is 0.1.
- The posterior probability of component $j$ is 0.3.
- The probability for observing a data sample $x$ given the total parameter set $\theta$ is 0.2.

Compute the likelihood for observing a data sample $x$ given component $j$ and the corresponding parameter vector. Round to two decimal places.

**Answer**

Start with Bayes' rule

$$P(j \mid x) = \frac{P(j)\, p(x \mid j)}{p(x \mid \theta)}.$$

Multiply by $p(x \mid \theta)$ and divide by $P(j)$ to isolate the unknown component likelihood:

$$p(x \mid j) = \frac{P(j \mid x)\, p(x \mid \theta)}{P(j)} = \frac{0.3 \cdot 0.2}{0.1} = \frac{0.06}{0.1} = \mathbf{0.60}.$$

**Question 43: Mixture of Gaussians: posterior of the first Gaussian**

The data generation process in a mixture of Gaussians model is as follows: first choose one Gaussian according to its weight in the mixture, then draw a sample from that Gaussian. You have fitted a mixture of two Gaussians to 1D data:

$$p(x) = \alpha_1 f(x; \mu_1, \sigma_1) + \alpha_2 f(x; \mu_2, \sigma_2),$$

with

$$\alpha_1 = 0.3, \quad \alpha_2 = 0.7, \quad \mu_1 = 0, \quad \sigma_1 = 1, \quad \mu_2 = 3, \quad \sigma_2 = 1.$$

Relevant densities are

$$f(1; 0, 1) = 0.242, \qquad f(1; 3, 1) = 0.054.$$

Calculate the probability that the datapoint $x = 1$ has been produced by the first Gaussian. Round to two decimal places.

**Answer**

The weighted density

$$\alpha_k f(x; \mu_k, \sigma_k)$$

is the joint contribution of component $k$ and observation $x$. The posterior responsibility of the first Gaussian is its contribution divided by the sum of the contributions of all components:

$$P(G_1 \mid x = 1) = \frac{\alpha_1 f(1; 0, 1)}{\alpha_1 f(1; 0, 1) + \alpha_2 f(1; 3, 1)} = \frac{0.3 \cdot 0.242}{0.3 \cdot 0.242 + 0.7 \cdot 0.054}$$

$$= \frac{0.0726}{0.0726 + 0.0378} = \frac{0.0726}{0.1104} \approx \mathbf{0.66}.$$

**Question 44: KL divergence for continuous uniform distributions — variant 2**

Assume $Y$ and $Z$ are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively. For $\theta_Y = e^4$ and $\theta_Z = e^{10}$, compute

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log\left(\frac{p_Y(x)}{p_Z(x)}\right) \mathrm{d}x.$$

Use the natural logarithm and round to two decimal places.

**Answer**

The support of $p_Y$ is $[0, \theta_Y]$. Because $\theta_Y = e^4 < e^{10} = \theta_Z$, it lies entirely inside the support of $p_Z$. On $[0, \theta_Y]$ the two uniform densities are $1/\theta_Y$ and $1/\theta_Z$, so their ratio is constant:

$$\begin{aligned} D_{\mathrm{KL}}(p_Y \| p_Z) &= \int_0^{\theta_Y} \frac{1}{\theta_Y} \ln\left(\frac{1/\theta_Y}{1/\theta_Z}\right) \mathrm{d}x \\ &= \ln\left(\frac{\theta_Z}{\theta_Y}\right) \underbrace{\int_0^{\theta_Y} \frac{1}{\theta_Y} \mathrm{d}x}_{=1} \\ &= \ln\left(\frac{e^{10}}{e^4}\right) = \ln(e^6) = \mathbf{6.00}. \end{aligned}$$

## Question 45: KL divergence for two discrete distributions

*Asked multiple times.*

You are given two probability distributions $p$ and $q$:

$$p(x=0) = 0.8, \quad p(x=1) = 0.2, \qquad q(x=0) = 0.5, \quad q(x=1) = 0.5.$$

Compute the KL divergence $D_{\mathrm{KL}}(p\|q)$. Use the natural logarithm and round to two decimal places.

**Answer**

For discrete distributions, replace the integral in the KL definition by a sum over all possible outcomes:

$$\begin{aligned} D_{\mathrm{KL}}(p\|q) &= \sum_x p(x) \ln \frac{p(x)}{q(x)} \\ &= p(0) \ln \frac{p(0)}{q(0)} + p(1) \ln \frac{p(1)}{q(1)} \\ &= 0.8 \ln \frac{0.8}{0.5} + 0.2 \ln \frac{0.2}{0.5} \\ &\approx 0.8(0.470) + 0.2(-0.916) \\ &= 0.376 - 0.183 \approx \mathbf{0.19}. \end{aligned}$$

The second outcome contributes a negative term, but the sum is nonnegative, as every KL divergence must be.

## Question 46: Clustering — version A

Which statements about Clustering are correct? Multiple answers are possible.

- ☐ In affinity propagation, the cluster centers are always data points.
- ☐ Clustering (or Cluster Analysis) is both a regression and a classification technique.
- ☐ In both $k$-means and fuzzy $k$-means, the membership distribution is always given by a binary random variable.
- ☐ In mixture models, an observation is assigned to the component with the largest prior distribution.
- ☐ Top-down clustering relies on building the minimal spanning tree and then in each step removes the largest of the remaining edges.

- ☐ In a mixture of Gaussians, the maximum likelihood and the maximum posterior always have the same value.
- ☐ The agglomerative hierarchical clustering algorithm repeatedly fuses the two maximally distant clusters.
- ☐ Similarity-based clustering does not require objects to be represented via feature vectors.

**Answer**

- ✓ **A.** Affinity propagation selects *exemplars* — actual data points — as cluster centers.
- × **B.** Clustering is unsupervised — it is neither regression nor classification (both are supervised).
- × **C.** *Fuzzy* $k$-means is exactly the variant with continuous (soft) memberships; only plain $k$-means is binary.
- × **D.** Assignment goes to the largest *posterior* (responsibility), which also accounts for how well the component explains the observation — not the prior alone.
- ✓ **E.** This is exactly the lecture's description of divisive (top-down) clustering: build the minimal spanning tree, remove the largest edge to split the data into two clusters, and recurse. (The resulting partitions coincide with single-linkage cuts, but the *procedure* — start with one cluster, split repeatedly — is top-down.)
- × **F.** MAP includes the prior term, so the two optima and their values differ in general.
- × **G.** Agglomerative clustering fuses the two *closest* clusters.
- ✓ **H.** It only needs pairwise similarities — objects never have to live in a vector space.

**Question 47: Clustering — version B**

Which statements about Clustering are correct? Multiple answers are possible.

- ☐ $k$-means is faster than MoG and not sensitive to initialization.
- ☐ Both $k$-means and MoG assign an observation to a single cluster only.
- ☐ The mixing coefficients in a Mixture of Gaussians model are represented by the posteriors.
- ☐ The maximum a posteriori estimation is a way to avoid near-zero variances and singular solutions in the maximum likelihood estimation of mixture models.
- ☐ Given an observation $x$, a component $j$, and a parameter vector $\theta$, the posterior $p(j \mid x, \theta)$ describes how likely it is that $x$ was produced by $j$.
- ☐ The EM algorithm for MoG tries to maximize the likelihood with respect to the cluster centers, the covariance matrices, and the mixing coefficients.
- ☐ The $k$-means algorithm is a simplification of the Mixture of Gaussians model.
- ☐ Clustering is an unsupervised technique that aims at finding separate regions of high data density called clusters.

**Answer**

- × **A.** Half-true trap: $k$-means *is* faster than MoG, but it is definitely sensitive to initialization.

✗ **B.** $k$-means does, but MoG assigns *soft* (probabilistic) memberships to all components.

✗ **C.** The mixing coefficients are the *priors* $\alpha_j$, not the posteriors.

✓ **D.** The prior in MAP regularizes the covariances away from collapsing onto single points (singular ML solutions).

✓ **E.** That is the definition of the responsibility $p(j \mid x, \theta)$.

✓ **F.** Means, covariances, and mixing coefficients are exactly the parameters EM optimizes.

✓ **G.** $k$-means is the hard-assignment, equal-spherical-covariance limit of MoG.

✓ **H.** Standard definition of clustering.

## Question 48: Hierarchical clustering: single linkage vs. complete linkage

You are given 8 datapoints in $\mathbb{R}^2$:

$$\{(0,1),(1,1),(2,1),(8,1),(0,-1),(1,-1),(2,-2),(8,-1)\}.$$

Agglomerative hierarchical clustering is run until only two clusters remain, once with single linkage and once with complete linkage. Two resulting clusterings are shown in the exam figure. Check the correct statements.

☐ Clustering 1 has been obtained by single linkage.

☐ Clustering 2 has been obtained by single linkage.

☐ Clustering 2 has been obtained by complete linkage.

☐ Clustering 1 has been obtained by complete linkage.

☐ One of the two clusterings could theoretically have been the result of either linkage function.

## Answer

✗ **A.** See D.

✓ **B.** Single linkage merges via the *closest* pair, so it chains the six left points into one elongated cluster and leaves the two distant points at $x = 8$ as the second cluster — the chained partition is clustering 2.

✗ **C.** See B.

✓ **D.** Complete linkage merges via the *farthest* pair, so it penalizes large cluster diameters and produces the compact partition — clustering 1.

✓ **E.** Correct: the linkage functions only differ once cluster diameters start to matter; for this dataset one of the two shown partitions can arise under either criterion.

## Question 49: Hierarchical clustering with average linkage on letters

*Asked multiple times.*

You are given the set of data points

$$x = \{a, c, e, h, i, m, n, s\},$$

where the similarity between two points $s(x, x')$ is defined as the distance between the ordered letters in the alphabet, for example $s(a, b) = 1$, $s(a, c) = 2$, and so on. Perform agglomerative

clustering using average linkage,

$$d_{\text{avg}}(A, B) = \frac{1}{n_A}\frac{1}{n_B}\sum_{a\in A}\sum_{b\in B} s(a, b),$$

and stop when the distance between clusters is greater than 7. How many clusters are formed in total, what is the maximum number of data points in any one cluster, and what is the minimum distance between any two of the formed clusters?

**Answer**

Alphabet positions: $a = 1$, $c = 3$, $e = 5$, $h = 8$, $i = 9$, $m = 13$, $n = 14$, $s = 19$. Merging the closest clusters step by step:

1. $\{h, i\}$ and $\{m, n\}$ at distance 1 each.
2. $\{a, c\}$ at distance 2.
3. $\{a, c\} + \{e\}$: $d_{\text{avg}} = \frac{4+2}{2} = 3$.
4. $\{h, i\} + \{m, n\}$: $d_{\text{avg}} = \frac{5+6+4+5}{4} = 5$.
5. Remaining distances: $d(\{a, c, e\}, \{h, i, m, n\}) = \frac{7+8+12+13+5+6+10+11+3+4+8+9}{12} = 8$ and $d(\{h, i, m, n\}, \{s\}) = \frac{11+10+6+5}{4} = 8$. The third distance is $d(\{a, c, e\}, \{s\}) = \frac{18+16+14}{3} = 16$. The smallest remaining inter-cluster distance is therefore $8 > 7$, so we **stop**.

Result: **3 clusters** $\{a, c, e\}$, $\{h, i, m, n\}$, $\{s\}$; the largest contains **4** points; the minimum distance between the formed clusters is **8**.

**Question 50: Biclustering**

Which statements about Biclustering are correct? Multiple answers are possible.

- ☐ A common biclustering method is the variance maximization method.
- ☐ FABIA is a generative model.
- ☐ FABIA defines biclusters as inner products.
- ☐ In FABIA, the likelihood is analytically intractable and requires, for example, a variational EM algorithm.
- ☐ Row and column elements must belong to exactly one bicluster or to no bicluster at all.
- ☐ Biclustering simultaneously clusters the rows and columns of a matrix.

**Answer**

- × **A.** Made-up name — not a biclustering method.
- ✓ **B.** FABIA is a generative model of the data matrix (factor analysis with sparse priors).
- × **C.** Near-miss: FABIA models a bicluster as an *outer* product $\lambda z^\top$ of a sparse loading and a sparse factor, not an inner product.
- ✓ **D.** The FABIA likelihood has no closed form; it is trained with variational EM.
- × **E.** Biclusters may overlap — rows and columns can belong to several biclusters or to none.
- ✓ **F.** That is the definition of biclustering.